GND
VCC

Pin header connections

J1
EPSM_ADDR

GND	1	2	VCC
B_CPU_A11	3	4	B_M2
B_CPU_A10	5	6	B_CPU_A12
B_CPU_A9	7	8	B_CPU_A13
B_CPU_A8	9	10	B_CPU_A14
B_CPU_A7	11	12	
B_CPU_A6	13	14	EPSM_A1
B_CPU_A5	15	16	EPSM_A0
B_CPU_A4	17	18	EPSM_CE3
B_CPU_A3	19	20	EPSM_CE2
B_CPU_A2	21	22	EPSM_CE1
B_CPU_A1	23	24	
B_CPU_A0	25	26	GND
B_CPU_RW	27	28	B_ROMSEL
	29	30	GND

J2
EPSM_DATA

GND	1	2	VCC
YM_CS	3	4	B_CPU_D7
YM_A0	5	6	B_CPU_D6
YM_A1	7	8	B_CPU_D5
YM_D7	9	10	B_CPU_D4
YM_D6	11	12	B_CPU_D3
YM_D5	13	14	B_CPU_D2
YM_D4	15	16	B_CPU_D1
YM_D3	17	18	B_CPU_D0
YM_D2	19	20	EPSM_A1
YM_D1	21	22	EPSM_A0
YM_D0	23	24	EPSM_CE3
GND	25	26	EPSM_CE2
GND	27	28	EPSM_CE1
VCC	29	30	GND

Defaults to $4016/$401C-F addr
To use different addressing, disconnect jumpers.

JP1 GND — EPSM_CE1
JP2 $401CFWCE — EPSM_CE2
JP3 VCC — EPSM_CE3
JP4 B_CPU_A2 — EPSM_A0
JP5 B_CPU_A1 — EPSM_A1

Famicom bus lines must be buffered to 5V CMOS!

$4016/$401C-F multiplexer

U9 74HC157
U13 74HC157
U14 74HC157

	U9	U13	U14
I0a	EPSM_CS	B_CPU_D6	B_CPU_D2
I1a	S_CS	S_CPU_D6	S_CPU_D2
I0b	EPSM_A1	B_CPU_D4	B_CPU_D0
I1b	S_A1	S_CPU_D4	S_CPU_D0
I0c	B_CPU_D7	B_CPU_D3	GND
I1c	S_CPU_D7	S_CPU_D3	GND
I0d	EPSM_A0	B_CPU_D5	B_CPU_D1
I1d	S_A0	S_CPU_D5	S_CPU_D1
S	$4016_S	$4016_S	$4016_S
E	GND	GND	GND
Za	YM_CS	YM_D6	YM_D2
Zb	YM_A1	YM_D4	YM_D0
Zc	YM_D7	YM_D3	
Zd	YM_A0	YM_D5	YM_D1

C12 0.1u
C15 0.1u
C17 0.1u

$4016 pulse delay control select

U10A 74HC132
U10B 74HC132
U10C 74HC132
U10D 74HC132
U10E 74HC132

$4016_P — R1 3K — C9 68p — GND
$4016_P — R2 1K — C10 68p — GND
$4016_P_DL — R3 1K — C11 68p — GND

U10A: 1 $4016_P, 2 $4016_P_DL, 3 $4016_P_DL
U10B: 4 $4016_P_DL, 5 S_CS, 6 S_CS
U10C: 8 S_RST, 9 $4016_P_DL, 10 $4016_P_DL
U10D: 11 $4016_P_DL, 12 $4016_P_DL, 13 $4016_P_DL
U10E: 14 VCC, 7 GND
C16 0.1u

74'138 address decoding

U1 74HC138
$40xx_W_A
CPU_A14 = 1
CPU_A12 = 0
CPU_A11 = 0
CPU_A13 = 0
CPU_RW = 0
M2 = 1
C1 0.1uF

U5 74HC138
$40xx_W_B
CPU_A6 = 0
CPU_A7 = 0
CPU_A8 = 0
CPU_A9 = 0
CPU_A10 = 0
ROMSEL = 1
C5 0.1uF

U2 74HC138
$4016_W
CPU_A1 = 1
CPU_A2 = 1
CPU_A0 = 0
CPU_A3 = 0
CPU_A5 = 0
CPU_A4 = 1
C2 0.1uF

U6 74HC138
$401C-F_W
CPU_A2 = 1
CPU_A3 = 1
CPU_A5 = 0
CPU_A4 = 1
C6 0.1uF

U8A 74HC32
U8D 74HC32
U8B 74HC32
U8C 74HC32
U8E 74HC32
C8 0.1uF

U11 74LVC1GU04
C13 0.1u

Alternative 74'688 address decoding

U15 74HC688
$40xx
CPU_A8 = 0
CPU_A9 = 0
CPU_A10 = 0
CPU_A11 = 0
CPU_A12 = 0
CPU_A13 = 0
CPU_A14 = 1
ROMSEL = 1
C18 0.1u

U16 74HC688
$401C-F W
CPU_A2 = 1
CPU_A3 = 1
CPU_A4 = 1
CPU_A5 = 0
CPU_A6 = 0
CPU_A7 = 0
CPU_RW = 0
M2 = 1
C19 0.1u

U17 74HC688
$4016
CPU_A0 = 0
CPU_A1 = 1
CPU_A2 = 1
CPU_A3 = 0
CPU_A4 = 1
CPU_A5 = 0
CPU_A6 = 0
CPU_A7 = 0
C20 0.1u

U18 74HCT1G02
C21 0.1u

U19 74HCT1G08
C22 0.1u

$4016 OUT1 data/address latch

U3 74HC174
U7 74HC174
C3 0.1u
C7 0.1u

U4A 74HC139
U4B 74HC139
U4C 74HC139
C4 0.1u

OUT1 latch reconstruction

U12 74LVC1G373
C14 0.1u

Some pinouts subject to change over time for ease of layout
EPSM Copyright © 2021–2026 Perkka
Copyright © 2022–2026 Persune

Sheet: /
File: EPSM-FC Decoder Daughterboard.kicad_sch

Title: EPSM-FC $4016/$401C-F Address Decoder Daughterboard

Size: A3	Date: 2026-11-04	**Rev: r.0.1.0**
KiCad E.D.A. 10.0.4		Id: 1/1